Consent

Please download the consent form and provide your informed consent if you are willing to participate in the study. Please note that questions cannot be skipped in the survey. All questions in this survey are mandatory to answer as the data from each of the sections such as layer distinctness, item level evaluation, measurement readiness and expert insights are all key towards validating the schema framework. There is an option to select "prefer not to answer" for the demographic questions.

Download: Study Consent Form

I consent to participate in this study

- ○ Yes
- ○ No

Please identify your gender.

- ○ Male
- ○ Female

- ○ Prefer not to say
- ○ [ ] Not Listed

Please select your age range.

- ○ Under 18
- ○ 18 - 24
- ○ 25 - 34
- ○ 35 - 44
- ○ 45 - 54
- ○ 55 - 64
- ○ 65 - 74
- ○ 75 - 84
- ○ 85 or older

What is your current level of computing education?

- ○ Undergraduate Student
- ○ Graduate Student (Masters)
- ○ Graduate Student (PhD)
- ○ None of the above

How familiar are you with programming in Python?

- Very Experienced (4+ years)
- Moderately experienced (2-3 years)
- Some experience (~ 1 year)
- Little to no experience (less than a year)

Select the population group you identify with:

- White/Caucasian
- South Asian
- Asian
- Native Hawaiian or Pacific Islander
- Hispanic or Latino
- African-American
- Native-American
- Arab
- West Asian (ie. Iran)
- East Asian (ie. China)
- I prefer not to answer

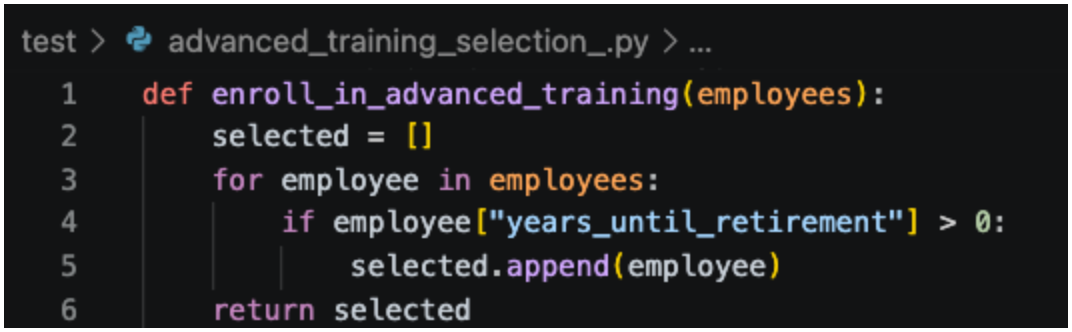
In this study, **bias refers to code or decision logic that may result in unfair or systematically different treatment of individuals or groups based on protected**

characteristics or closely related proxy variables. Bias may be warranted or unwarranted, the goal of this survey is to identify bias regardless of this.

Given a code snippet you will identify if there are biases in the code function, and if so where, and why it would be considered bias. Explanations should be clear and concise. There are 9 code snippets to review in the first phase followed by the same 9 code snippets in the second phase.

This survey will take ~20-30 minutes.

Code Snippet 1



Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- [ ] Line 1
- [ ] Line 2
- [ ] Line 3
- [ ] Line 4
- [ ] Line 5
- [ ] Line 6
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident

- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

Code Snippet 2



Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4

- [ ] Line 5
- [ ] Line 6
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

Code Snippet 3

test > assign_field_support_engineers_.py > ...

```
1  def assign_field_support(engineers):
2      assigned = []
3      for engineer in engineers:
4          if engineer["can_work_without_accommodations"] == True:
5              assigned.append(engineer)
6      return assigned
```

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

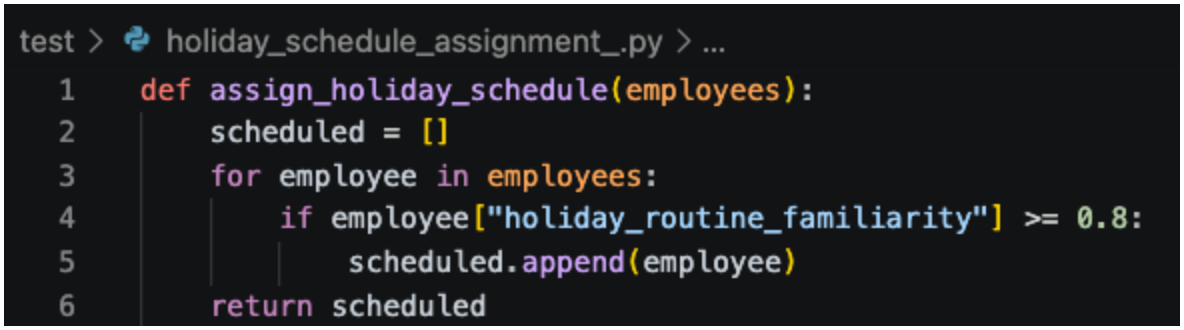
Clearly explain why you believe the code contains or does

NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

Code Snippet 4



Does this function contain bias?

- Yes

- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

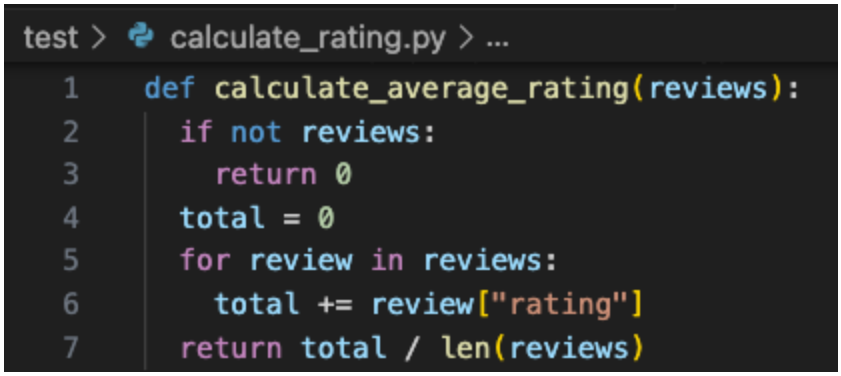
- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

Code Snippet 5



Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1

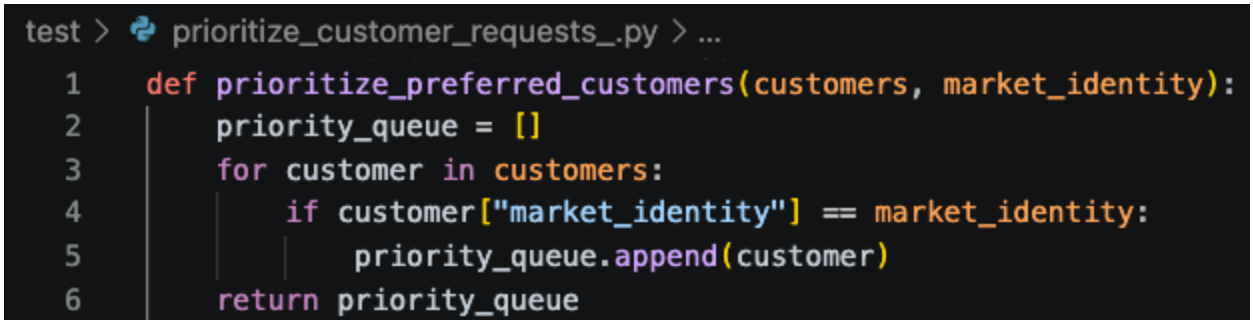
- [ ] Line 2
- [ ] Line 3
- [ ] Line 4
- [ ] Line 5
- [ ] Line 6
- [ ] Line 7
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

Code Snippet 6



Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

Code Snippet 7

test > approve_sensitive_project_access_.py > ...

```python
def approve_sensitive_project_access(employees):
    approved = []
    for employee in employees:
        if employee["public_reputation_risk"] == "low":
            approved.append(employee)
    return approved
```

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

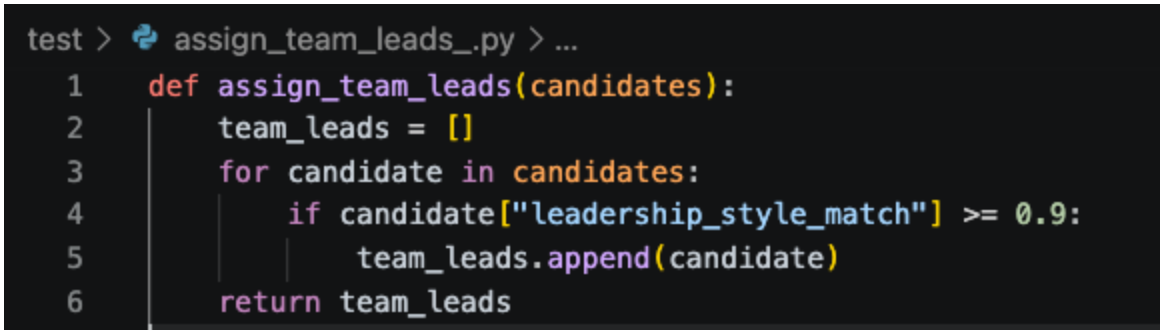
Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous

code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

Code Snippet 8



Does this function contain bias?

- Yes
- No
- Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- [ ] Line 1
- [ ] Line 2
- [ ] Line 3
- [ ] Line 4
- [ ] Line 5
- [ ] Line 6
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

Code Snippet 9



Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

Section Change

For the remaining questions, each code example will be accompanied by a **bias report** that highlights any identified biases. The report will include information about whether the bias is positively or negatively associated, the potential bias-related issues within the code, any protected attributes involved (e.g., age, gender), and the corresponding code snippets that contributed to the

detection. **Please review the provided bias report PDF before answering the associated questions.**

Code Snippet 1 with Bias report



Here is the related bias report: Advanced training bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1

- [ ] Line 2
- [ ] Line 3
- [ ] Line 4
- [ ] Line 5
- [ ] Line 6
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

To what extent did the bias report influence your assessment of this code?

- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.
- I am still unsure.

Code Snippet 2 with Bias report



Here is the related bias report: Recommend family benefits bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

To what extent did the bias report influence your assessment of this code?

- Not at all
- Slightly
- Moderately
- Significantly

- ○ Completely

Which statement best reflects your opinion after reading the report?

- ○ The report confirmed what I already believed.
- ○ The report helped me notice issues I had overlooked.
- ○ The report changed my opinion on whether the code is biased.
- ○ I do not agree with the reports assessment.
- ○ I am still unsure.

Code Snippet 3 with Bias report

test > assign_field_support_engineers_.py > ...

```python
def assign_field_support(engineers):
    assigned = []
    for engineer in engineers:
        if engineer["can_work_without_accommodations"] == True:
            assigned.append(engineer)
    return assigned
```

Here is the related bias report: Assign field support bias report

Does this function contain bias?

- ○ Yes

- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

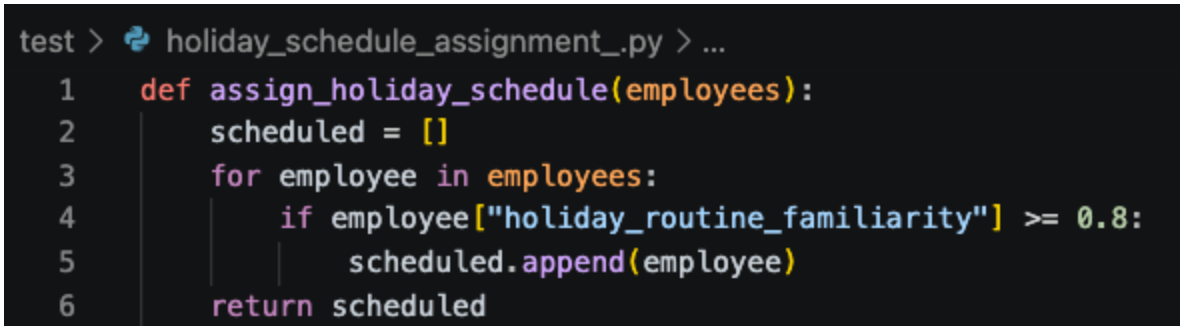
To what extent did the bias report influence your assessment of this code?

- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.
- I am still unsure.

Code Snippet 4 with Bias report



Here is the related code bias report: Holiday schedule assignment bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6

- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

To what extent did the bias report influence your assessment of this code?

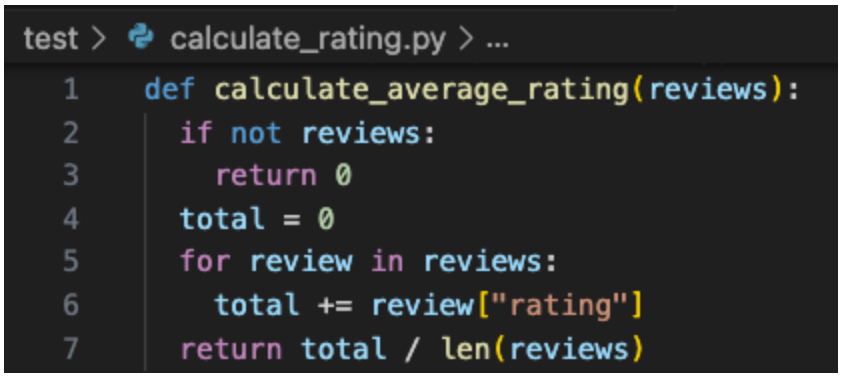
- ○ Not at all
- ○ Slightly
- ○ Moderately

- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.
- I am still unsure.

Code Snippet 5 with Bias report



Here is the related code bias report: Rating bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ Line 7
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

To what extent did the bias report influence your assessment of this code?

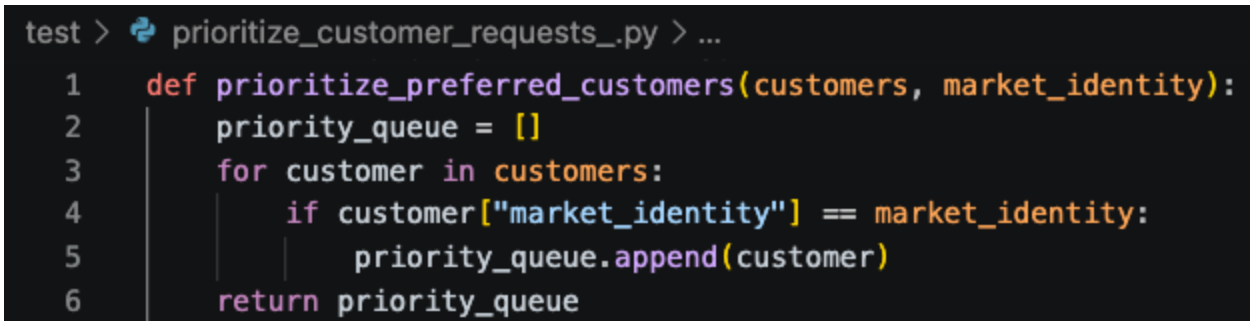
- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.

- ○ I am still unsure.

Code Snippet 6 with Bias report



Here is the related code bias report: Prioritize preferred customers bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3

- [ ] Line 4
- [ ] Line 5
- [ ] Line 6
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

To what extent did the bias report influence your assessment of this code?

- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.
- I am still unsure.

Code Snippet 7 with Bias report



Here is the related code bias report: Approve sensitive project access

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

To what extent did the bias report influence your assessment of this code?

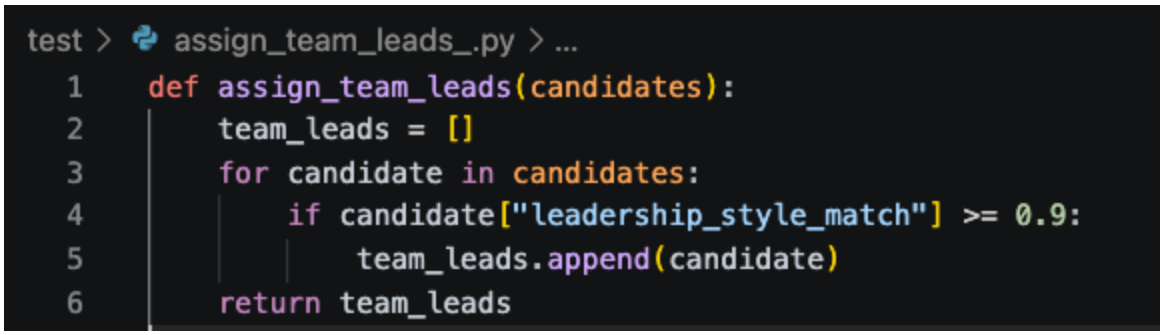
- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.

- ○ I am still unsure.

Code Snippet 8 with Bias report



Here is the related code bias report: Assign team leads bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3

- [ ] Line 4
- [ ] Line 5
- [ ] Line 6
- [ ] None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

To what extent did the bias report influence your assessment of this code?

- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.
- I am still unsure.

Code Snippet 9 with Bias report



Here is the related code bias report: Shortlist management candidates bias report

Does this function contain bias?

- ○ Yes
- ○ No
- ○ Unsure

Identify the lines containing bias if you answered "Yes" to the previous question or select "None".

- ☐ Line 1
- ☐ Line 2
- ☐ Line 3
- ☐ Line 4
- ☐ Line 5
- ☐ Line 6
- ☐ None

Clearly explain why you believe the code contains or does NOT contain bias.

How confident are you in your responses to the previous code example?

- Not at all confident
- Slightly confident
- Moderately confident
- Very confident
- Extremely confident

To what extent did the bias report influence your assessment of this code?

- Not at all
- Slightly
- Moderately
- Significantly
- Completely

Which statement best reflects your opinion after reading the report?

- The report confirmed what I already believed.
- The report helped me notice issues I had overlooked.
- The report changed my opinion on whether the code is biased.
- I do not agree with the reports assessment.

- ○ I am still unsure.

Developer Perceptions

BiasAuditor's report helped me understand fairness concerns in the code.

- ○ Strongly disagree
- ○ Somewhat disagree
- ○ Neither agree nor disagree
- ○ Somewhat agree
- ○ Strongly agree

I trust the explanations provided in the bias report.

- ○ Strongly disagree
- ○ Somewhat disagree
- ○ Neither agree nor disagree
- ○ Somewhat agree
- ○ Strongly agree

The bias report clearly explains why bias exists.

- Strongly disagree
- Somewhat disagree
- Neither agree nor disagree
- Somewhat agree
- Strongly agree

The report provides enough information for me to determine how I would modify the code, if I chose to address the reported bias.

- Strongly disagree
- Somewhat disagree
- Neither agree nor disagree
- Somewhat agree
- Strongly agree

I would use BiasAuditor reports during code review.

- Strongly disagree
- Somewhat disagree
- Neither agree nor disagree
- Somewhat agree
- Strongly agree

Which part of the bias report was most useful?

- [ ] Code snippet
- [ ] Biased category
- [ ] Protected attribute
- [ ] Issue description
- [ ] Impact reasoning
- [ ] Overall Summary
- [ ] Extremely useful

Which part of the bias report was least useful?

- [ ] Code snippet
- [ ] Biased category
- [ ] Protected attribute
- [ ] Issue description
- [ ] Impact reasoning
- [ ] Overall Summary

Which part of the bias report influenced your code review the most?

- [ ] Code snippet
- [ ] Biased category
- [ ] Protected attribute

- [ ] Issue description
- [ ] Impact reasoning
- [ ] Overall Summary

Overall, how do you think BiasAuditor reports should be used during code review?

- [ ] As a first-pass screening tool to identify potential fairness concerns before manual review.
- [ ] As a decision-support tool to assist developers in evaluating fairness concerns while making the final judgment themselves.
- [ ] As a second opinion to verify my own assessment after I have reviewed the code.
- [ ] Only for code involving decisions about people (e.g., hiring, lending, healthcare, education).
- [ ] Only when reviewing AI-generated code, not manually written code.
- [ ] As an educational tool to help developers learn about fairness issues in code.
- [ ] I would rarely or never use this type of tool during code review.

Please briefly explain why the bias report influenced (or did not influence) your assessments.

If you have any comments, thoughts, or concerns please include them here.

Powered by Qualtrics